33

# BIOLOGY HSSC-I

## SECTION – A (Marks 17)

Time allowed: 25 Minutes

**Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.**

*Do not use lead pencil.*

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

| Version No. | | | | |
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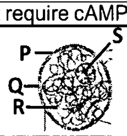
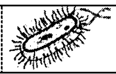
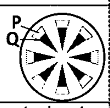
ROLL NUMBER

**Answer Sheet No.** ____________

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔ **Invigilator Sign.** ______________

**Fill the relevant bubble against each question according to curriculum:** **Candidate Sign.** ______________

| | Question | A | B | C | D | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| 1. | In tRNA molecule, the main function of middle loop is to: | Recognize enzyme activity | Bind codon on mRNA | Recognize binding site on ribosome | Attach amino acid to CCA | | | | |
| 2. | Prostaglandins are derivatives of: | Amino acids | Steroids | Archidonic acid | Nucleic acids | | | | |
| 3. | What is correct for protein signaling?<br>I Water soluble<br>II Binds receptor on plasma membrane<br>III Lipophilic IV Do not require cAMP | I & II | II & III | III & IV | I & IV | | | | |
| 4. | Which labelled part in nucleus is responsible for ribosome synthesis?   | P | Q | R | S | | | | |
| 5. | Which enzymes break specific covalent bonds without hydrolysis? | Oxidoreductases | Hydrolases | Transferases | Lyases | | | | |
| 6. | Viroids cause: | Hepatitis A | Hepatitis B | Hepatitis C | Hepatitis D | | | | |
| 7. | Type of flagella arrangement shown in bacteria is:  | Lophotrichous | Peritrichous | Amphitrichous | Monotrichous | | | | |
| 8. | Genetic material moves from donor to recipient bacterium in: | Binary fission | Conjugation | Transduction | Transformation | | | | |
| 9. | Mutualistic fungi which grows between root cells of plants and does not penetrate cells is: | Lichens | Endomycorrhiza | Ectomycorrhiza | Epiphytes | | | | |
| 10. | Growth movements shown by climbing plants with a supporting structure is: | Phototropism | Geotropism | Thigmotropism | Chemotropism | | | | |
| 11. | During unwinding of DNA double helix, the hydrogen bonds between base pairs are broken by enzyme: | Helicase | Primase | Polymerase | Gyrase | | | | |
| 12. | Radial symmetry and water vascular system are found in phylum: | Hemichordata | Echinodermata | Chordata | Vertebrata | | | | |
| 13. | Uterine lining in female starts breaking due to less concentration of: | Progesterone | Estrogen | Luteinizing hormone | FSH | | | | |
| 14. | A forest was completely destroyed by fire. After about hundred year, a new forest was evolved. The type of succession shown is: | Primary and xerarch | Secondary and xerarch | Primary and hydrarch | Secondary and hydrarch | | | | |
| 15. | Holandric traits show all of the following characteristics **EXCEPT**: | Sons of affected father are affected | Show Y-linked inheritance | Porcupine man is an example | Daughters of affected father are carriers | | | | |
| 16. | In TS of a dicot stem, which substances are expected to pass through labelled parts P and Q respectively?   | Water and sucrose | Sucrose and amino acids | Sucrose and water | Water and amino acids | | | | |
| 17. | The structures which help bacteria to survive under unfavorable conditions include: | Flagella and pili | Pili and akinetes | Endospores and plasmid | Flagella and heterocyst | | | | |

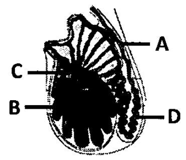
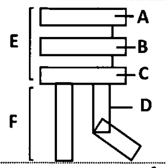
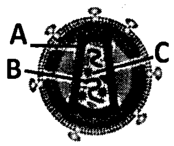
——1HA-I 25006 (D)——

# BIOLOGY HSSC–I

**Time allowed: 2:35 Hours** **Total Marks Sections B and C: 68**

## *SECTION – B (Marks 42)*

**Q. 2 Answers the following parts briefly.** **(14 x 3 = 42)**

<table>
<tr><td>(i)</td><td>The diagram shows the section through testis:<br>a) Identify the parts A & B.<br>b) Write functions of C & D.<br>c) What is the function of inhibin hormone in males?<br></td><td>1x3</td><td>OR</td><td>Write the major points of theory of natural selection. (Any three)</td><td>03</td></tr>
<tr><td>(ii)</td><td>Compare any three adaptations for osmotic adjustments in hydrophytes and mesophytes.</td><td>03</td><td>OR</td><td>Why post–transcriptional modifications occur in eukaryotic mRNA? Summarize the changes which occur during mRNA processing.</td><td>1+2</td></tr>
<tr><td>(iii)</td><td>Write about the structure of prions. Enlist two diseases caused by them.</td><td>2+1</td><td>OR</td><td>Write any three distinguishing characteristics of vascular plants.</td><td>03</td></tr>
<tr><td>(iv)</td><td>What are the homologous organs? Write an example. Which type of evolution is caused by them?</td><td>1x3</td><td>OR</td><td>Write any three chemical methods with their mechanisms used to control bacteria.</td><td>1x3</td></tr>
<tr><td>(v)</td><td>Complete the table:<table><tr><th></th><th>Group</th><th>Character</th><th>Example</th></tr><tr><td>a)</td><td>Bryophytes</td><td></td><td></td></tr><tr><td>b)</td><td></td><td>Seedless plants</td><td></td></tr><tr><td>c)</td><td></td><td></td><td>Rose</td></tr></table></td><td>03</td><td>OR</td><td>Name the type of pattern of sex determination in humans. Also write mechanism of sex determination in man.</td><td>1+2</td></tr>
<tr><td>(vi)</td><td>Write the difference between leaf, seed and root of monocots and dicots.</td><td>03</td><td>OR</td><td>Write down any three roles of smooth endoplasmic reticulum in a cell.</td><td>03</td></tr>
<tr><td>(vii)</td><td>Peroxisomes are more abundant in liver cells. Justify.</td><td>03</td><td>OR</td><td>Outline the process of non-cyclic photophosphorylation with the help of a labelled flow sheet.</td><td>03</td></tr>
<tr><td>(viii)</td><td>The diagram shows molecule of Lecithin:<br>a) Identify parts A, B, C & D.<br>b) Mention the polarity of E & F.<br></td><td>2+1</td><td>OR</td><td>Following is the diagram of HIV:<br>a) Identify the parts A, B & C.<br>b) Write the functions of protein gp40, gp120 and enzyme integrase in HIV.<br></td><td>03</td></tr>
<tr><td>(ix)</td><td>Differentiate between absorption spectrum of Chlorophyll– a, b and Carotenoids.</td><td>03</td><td>OR</td><td>How can a virus survive under unfavorable conditions outside the host?</td><td>03</td></tr>
<tr><td>(x)</td><td>Justify that O $^{-ve}$ persons are universal donors and AB $^{+ve}$ are universal recipients of blood.</td><td>03</td><td>OR</td><td>How is photosynthesis in Cyanobacteria different from Bacteria? ( any three differences)</td><td>03</td></tr>
<tr><td>(xi)</td><td>How are Starch and Glycogen different from each other although both are polysaccharides?</td><td>03</td><td>OR</td><td>Why Lysosomes are also called suicidal bags?</td><td>03</td></tr>
<tr><td>(xii)</td><td>Enlist any three fungal diseases in humans with one symptom each.</td><td>03</td><td>OR</td><td>Name the types of chromosomes on the basis of position of centromere mentioning its location in each type. (Any three)</td><td>03</td></tr>
<tr><td>(xiii)</td><td>Write about the structure and function of each of the three layers of uterus.</td><td>03</td><td>OR</td><td>How evolution of Giraffe neck can be explained on the basis of Lamarckism?</td><td>03</td></tr>
<tr><td>(xiv)</td><td>Write down the sources of mutagens with their examples. (Any three)</td><td>03</td><td>OR</td><td>Name any three main greenhouse gases and their sources of emission.</td><td>03</td></tr>
</table>

## *SECTION – C (Marks 26)*

**Attempt the following questions.**

| | | | | | |
|---|---|---|---|---|---|
| Q.3 | Explain Meselson–Stahl experiment and results to prove that DNA replication is semi conservative. Also draw the diagram. | 3+2 +2 | OR | Explain how the movement of protons across mitochondrial membrane is involved in ATP synthesis. Also draw diagram showing movement of electrons. | 5+2 |
| Q.4 | Describe Nitrogen Fixation, Ammonification and Nitrification stages involved in Nitrogen cycle. Also draw labelled Nitrogen cycle flow sheet. | 5+2 | OR | Explain mechanism of enzyme action through ‘induced fit’ and ‘lock and key’ models. | 4+3 |
| Q.5 | Epistasis is different from dominance. Explain with the example of coat color in *Labrador retriever*. | 06 | OR | Identify and explain the properties of water which enable it to:<br>a) Act as temperature stabilizer<br>b) Help organisms to live in warm environment<br>c) Help organisms to live under ice | 2x3 |
| Q.6 | Describe the components of theory explaining the ascent of sap in plants. Also draw the diagram. | 4+2 | OR | Write three distinguishing characteristics and one example of each of the phylum Porifera, Platyhelminthes and Annelida. | 2x3 |